

A Quantitative SNR Analysis of LFM signals in the Linear Canonical Transform Domain with Gaussian Windows

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Abstract—In this paper, the relationship between the linear canonical transform (LCT) and the Wigner-Ville distribution (WVD) is analyzed and proved. Based on this relationship, we present a quantitative signal-to-noise ratio (SNR) analysis of linear frequency modulated (LFM) signals in the LCT domain with Gaussian windows. The 3dB SNR definition is used. The SNR in LCT domain is compared with the SNRs in time domain, the short-time Fourier transform (STFT) domain and the pseudo-WVD domain, respectively, and come to a conclusion that the SNR in the LCT domain can achieve a significantly higher level than the SNRs in the other three domains. Some simulation results are presented to confirm the theoretical results.

Keywords—Linear canonical transform; SNR analysis; LFM signal; Gaussian window

I. INTRODUCTION

It's knowing that the time-frequency representations (TFRs), such as the short-time Fourier transform (STFT) and the Wigner-Ville distribution (WVD), can concentrate linear frequency modulated (LFM) signals while spreading noise in the time-frequency domain [1]. Therefore, the TFRs has the ability to improve the regional signal-to-noise ratio (SNR). Some quantitative SNR Analysis in the STFT and the WVD domain, have been studied in [2-5].

Another method that can concentrate LFM signals is the linear canonical transform (LCT). LCT is a linear integral transform which provides a generalization of both the Fourier transform (FT) and the fractional Fourier transform (FRFT) [6]. In recent years, the LCT is becoming one of the hottest research topics [6-11]. LCT with parameter $A = (a, b, c, d)$ of a signal $f(t)$ is defined as [6]:

$$L_f^A(u) = \begin{cases} \int_{-\infty}^{\infty} f(t)K_A(u, t)dt, & b \neq 0, \\ \sqrt{d} \exp\left(j\frac{cd u^2}{2}\right) f(du), & b = 0, \end{cases} \quad (1)$$

The parameters (a, b, c, d) are real numbers which satisfying $ad - bc = 1$, and $K_A(u, t)$ is the transformation kernel defined as:

$$K_A(u, t) = \frac{1}{\sqrt{j2\pi b}} \exp\left(j\frac{at^2}{2b} - j\frac{ut}{b} + j\frac{du^2}{2b}\right) \quad (2)$$

To simplify the expressions, the integrals without limits in the rest of the paper all imply the integrations are from $-\infty$ to ∞ .

When $A = (0, 1, -1, 0)$, LCT becomes conventional Fourier transform multiplied by factor $\sqrt{-j}$ as

$$\begin{aligned} L_f^{(0,1,-1,0)}(u) &= \frac{1}{\sqrt{j2\pi}} \int f(t) \exp(-jut) dt \\ &= \sqrt{-j} FT(f(t))(u). \end{aligned} \quad (3)$$

When $b = 0$, LCT just becomes a chirp multiplication operation, thus, in this paper, only $b \neq 0$ is considered.

For non-stationary signals, the variable t in the LCT definition (1) need to be truncated by a short window. Therefore, in this paper, the windowed version of the LCT, defined as

$$L_f^A(t, u) = \int f(t + \tau) h^*(\tau) K_A(u, \tau) d\tau, \quad (4)$$

is employed. The $h(\tau)$ in (4) is the window function.

In this paper, we present a quantitative SNR analysis of LFM signals in the LCT domain defined in (4), and Gaussian window function, defined as

$$h(\tau) = (\alpha/\pi)^{1/4} \exp\left(-\frac{\alpha\tau^2}{2}\right), \quad \alpha > 0, \quad (5)$$

is used. The α in (5) is the parameter which controls the width of the window.

II. SNR ANALYSIS

A. New SNR Definition

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For a given signal distorted by noise, the conventional SNR means the ratio of the mean power (over the whole domain) of the signal over the mean power of the noise. This SNR definition is not appropriate for LFM signals in the LCT domain, because these signals are non-stationary in this transform domain, so it's not proper to take over the whole time domain to get the mean power. In this paper, the 3dB SNR defined in [4] is used because it is suitable for LFM signals in LCT domain as well as in the time and frequency domains.

For an original signal $s[n]$ distorted by an additive noise $\eta[n]$, let

$$y[n] = s[n] + \eta[n], \quad 0 \leq n \leq N-1 \quad (6)$$

$\eta[n]$ is a noise with mean zero and variance σ^2 . The variable n in $y[n]$, $s[n]$ and $\eta[n]$ means these signals are in a same random domain.

For the signal $s[n]$ of length N , $0 \leq n \leq N-1$, let

$$\chi \triangleq \{n: 0 \leq n \leq N-1\}$$

$$\text{and } |x[n]|^2 \geq 0.5 \max_{0 \leq n \leq N-1} |x[n]|^2 \quad (7)$$

where the number 0.5 comes from the common 3-dB bandwidth definition in communications[4]. Then, the 3 dB SNR is defined as

$$SNR^{3db} \triangleq \frac{\sum_{n \in \chi} |x[n]|^2}{|\chi| \sigma^2} \quad (8)$$

where $|\chi|$ denotes the cardinality of the set χ .

B. Relationship between LCT and WVD

References [12] have researched the relationship between the WVD, which defined as [3]

$$WVD_f(t, \omega) = \int f\left(t + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) \exp(-j\omega\tau) d\tau, \quad (9)$$

and the local polynomial Fourier transform (LPFT), then used this relationship to analyze the 3dB SNR in the LPFT domain. Based on the same thought, the relationship between the WVD and the LCT defined in (4), can be expressed as

$$|L_s^A(t, u)|^2 = \frac{1}{4\pi^2 b} \iint WVD_s\left(t - x, \frac{u + ax - by}{b}\right) \cdot WVD_h(x, y) dx dy. \quad (10)$$

We prove this equality before we use it to analyze the 3dB SNR in the LCT domain. Assume $h(t)$ is a real even window function, then the equality (10) can be proved as follows:

$$\begin{aligned} & \frac{1}{4\pi^2 b} \iint WVD_s\left(t - x, \frac{u + ax - by}{b}\right) \cdot WVD_h(x, y) dx dy \\ &= \frac{1}{4\pi^2 b} \int \int \int \int s\left(t - x + \frac{m}{2}\right) s^*\left(t - x - \frac{m}{2}\right) \\ & \quad \cdot \exp\left[-j\left(\frac{u + ax - by}{b}\right)m\right] \cdot h\left(x + \frac{n}{2}\right) h^*\left(x - \frac{n}{2}\right) \end{aligned}$$

$$\begin{aligned} & \cdot \exp[-jyn] dmdndxdy \\ &= \frac{1}{2\pi b} \int \int \int s\left(t - x + \frac{m}{2}\right) s^*\left(t - x - \frac{m}{2}\right) \\ & \quad \cdot \exp\left[-j\left(\frac{u + ax}{b}\right)m\right] \cdot h\left(x + \frac{n}{2}\right) h^*\left(x - \frac{n}{2}\right) \\ & \quad \cdot \delta(n - m) dmdndxdy \\ &= \frac{1}{\sqrt{j2\pi b}} \cdot \frac{1}{\sqrt{-j2\pi b}} \cdot \exp\left(j\frac{d}{2b}\right) u^2 \\ & \quad \cdot \exp\left(-j\frac{d}{2b}\right) u^2 \\ & \quad \cdot \int \int s\left(t - x + \frac{n}{2}\right) s^*\left(t - x - \frac{n}{2}\right) \\ & \quad \cdot \exp\left[-j\left(\frac{u + ax}{b}\right)n\right] \cdot h\left(x + \frac{n}{2}\right) h^*\left(x - \frac{n}{2}\right) dndx \end{aligned} \quad (11)$$

Let $p = x - n/2$ and $q = x + n/2$, we have $x = (p+q)/2$ and $n = q - p$, therefore, the Jacobi determinant is

$$J = \begin{vmatrix} \frac{\partial x}{\partial p} & \frac{\partial x}{\partial q} \\ \frac{\partial n}{\partial p} & \frac{\partial n}{\partial q} \end{vmatrix} = 1. \quad (12)$$

Then, with $dndx = |J|dpdq$, (11) becomes

$$\begin{aligned} & \frac{1}{\sqrt{j2\pi b}} \cdot \int s(t - p) h^*(p) e^{j\left(\frac{u}{b}p + \frac{a}{2b}p^2 + \frac{d}{2b}u^2\right)} dp \\ & \cdot \left[\frac{1}{\sqrt{j2\pi b}} \cdot \int s(t - q) h^*(q) e^{j\left(\frac{u}{b}q + \frac{a}{2b}q^2 + \frac{d}{2b}u^2\right)} dq \right]^* \\ &= \int s(t + \tau) h^*(\tau) K_A(u, \tau) d\tau \\ & \quad \cdot \left(\int s(t + \tau) h^*(\tau) K_A(u, \tau) d\tau \right)^* \\ &= |L_s^A(t, u)|^2. \end{aligned}$$

After equality (10) proved, we can use this relationship between the WVD and the LCT to analyze the SNR of LFM signals in the LCT domain.

C. Quantitative SNR Analysis

As a general situation, a continuous-time LFM signal model with noise can be expressed as

$$y(t) = s(t) + \eta(t) = A \exp\left[j\left(a_1 t + \frac{b_1}{2} t^2\right)\right] + \eta(t), \quad (13)$$

where A is the signal amplitude, a_1 is the initial frequency, b_1 is the chirp rate of the signal, and $\eta(t)$ is the additive white Gaussian noise. The correlation function of $\eta(t)$ is defined as

$$R_\eta(t, \tau) = E[\eta(\tau)\eta^*(\tau)] = \sigma^2 \delta(t - \tau). \quad (14)$$

The LCT is a linear transformation, therefore, the LCTs of the signal $s(t)$ and the noise $\eta(t)$ can be considered separately.

According to [2], the WVD of the signal $s(t)$ was

$$WVD_s(t, \omega) = 2\pi A^2 \delta(\omega - b_1 t - a_1). \quad (15)$$

According to [13], the WVD of the Gaussian window function was

$$WVD_h(t, \omega) = 2 \exp \left[- \left(\alpha t^2 + \frac{1}{\alpha} \omega^2 \right) \right]. \quad (16)$$

Using (10), (15) and (16), the square of $|L_s^A(t, u)|$ becomes

$$\begin{aligned} |L_s^A(t, u)|^2 &= \frac{A^2}{\pi b} \iint \delta \left[y - \left(\frac{u + ax}{b} - b_1 t + b_1 x - a_1 \right) \right] \\ &\quad \cdot \exp \left[- \left(\alpha x^2 + \frac{1}{\alpha} y^2 \right) \right] dx dy \\ &= \frac{A^2}{\pi b} \int \exp \left\{ - \left[\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2 \right] x^2 \right. \\ &\quad \left. - \frac{2}{\alpha} \left(\frac{a}{b} + b_1 \right) \left(\frac{u}{b} - b_1 t - a_1 \right) x \right. \\ &\quad \left. + \frac{1}{\alpha} \left(\frac{u}{b} - b_1 t - a_1 \right)^2 \right\} dx \end{aligned} \quad (17)$$

Using integral formula [13]

$$\int_{-\infty}^{\infty} \exp(-Ax^2 \pm 2Bx + C) dx = \sqrt{\frac{\pi}{A}} \exp \left(-\frac{AC - B^2}{A} \right) \quad (18)$$

We have

$$\begin{aligned} |L_s^A(t, u)|^2 &= \frac{A^2}{\sqrt{\pi b}} \cdot \frac{1}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}} \\ &\quad \cdot \exp \left\{ - \frac{\left(\frac{u}{b} - b_1 t - a_1 \right)^2}{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2} \right\}. \end{aligned} \quad (19)$$

Thus, the maximum of $|L_s^A(t, u)|^2$ is

$$\max_{(t,u)} |L_s^A(t, u)|^2 = \frac{A^2}{\sqrt{\pi b}} \cdot \frac{1}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}}, \quad (20)$$

Which achieved at $u = b(b_1 t + a_1)$.

According to 3dB SNR definition (8), the 3dB mean power of the original signal is

$$\text{mean}_{(t,u) \in S} |L_s^A(t, u)|^2, \quad (21)$$

where

$$S = \left\{ (t, u) : |L_s^A(t, u)|^2 > \frac{A^2}{\sqrt{\pi b}} \cdot \frac{0.5}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}} \right\}. \quad (22)$$

Using (19), we have

$$S = \left\{ (t, u) : \left(\frac{u}{b} - b_1 t - a_1 \right)^2 < \left(\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2 \right) \ln 2 \right\}. \quad (23)$$

Thus, (21) becomes

$$\begin{aligned} \text{mean}_{(t,u) \in S} |L_s^A(t, u)|^2 &= \frac{A^2}{\sqrt{\pi b}} \cdot \frac{1}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}} \cdot \frac{1}{\sqrt{\ln 2}} \int_0^{\sqrt{\ln 2}} \exp(-u^2) du. \end{aligned} \quad (24)$$

Since [2]

$$\frac{1}{\sqrt{\ln 2}} \int_0^{\sqrt{\ln 2}} \exp(-u^2) du \approx 0.8, \quad (25)$$

we have

$$\text{mean}_{(t,u) \in S} |L_s^A(t, u)|^2 = \frac{A^2}{\sqrt{\pi b}} \cdot \frac{0.8}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}}. \quad (26)$$

Then, we calculated the 3dB mean power of the noise. Since the noise $\eta(t)$ is stationary, we have

$$\begin{aligned} E |L_\eta^A(t, u)|^2 &= E \left| \int \eta(t + \tau) h^*(\tau) K_A(u, \tau) d\tau \right|^2 \\ &= E \left| \int \frac{1}{\sqrt{j2\pi b}} \eta(t + \tau) h^*(\tau) e^{j \left(\frac{at^2}{2b} - j \frac{ut}{b} + j \frac{du^2}{2b} \right)} d\tau \right|^2 \\ &= \frac{\sigma^2}{2\pi b}. \end{aligned} \quad (27)$$

Thus, the 3dB SNR in the LCT domain is

$$SNR_{LCT}^{3dB} = \frac{1.6\sqrt{\pi}}{\sqrt{\alpha + \frac{1}{\alpha} \left(\frac{a}{b} + b_1 \right)^2}} SNR_t, \quad (28)$$

where $SNR_t = A^2/\sigma^2$ is the conventional defined SNR.

D. Comparisons

From (28), we can clearly see that when the LCT parameters (a, b, c, d) are given, the maximum of the SNR_{LCT}^{3dB} is reached when the Gaussian parameter $\alpha = |a/b + b_1|$, and the maximum is

$$SNR_{LCT}^{max} = \max_{\alpha} (SNR_{LCT}^{3dB}) = \frac{0.8\sqrt{2\pi}}{\sqrt{\left| \frac{a}{b} + b_1 \right|}} SNR_t. \quad (29)$$

When $|a/b + b_1| < 1.28\pi$, SNR_{LCT}^{3dB} is larger than SNR_t . When $-a/b = b_1$, the ratio between SNR_{LCT}^{max} and SNR_t becomes infinity, which means the 3dB SNR of LFM signals in the LCT domain can achieve a significantly higher level than that in the time domain.

We can also see when the Gaussian parameter α is given, the maximum of SNR_{LCT}^{max} is reached when $-a/b = b_1$, which means the maximum is not affected by the parameter d , and the maximum is

$$SNR_{LCT}^{max} = \max_{\alpha} (SNR_{LCT}^{3dB}) = \frac{1.6\sqrt{\pi}}{\sqrt{\alpha}} SNR_t. \quad (30)$$

When $\alpha < 2.56\pi$, SNR_{LCT}^{3dB} is larger than SNR_t . When $\alpha = 0$, the ratio between SNR_{LCT}^{max} and SNR_t also becomes infinity.

According to [2], the maximum 3dB SNR in the STFT domain is

$$\max_{\alpha'}(SNR_{STFT}^{3dB}) = \frac{0.8\sqrt{2\pi}}{\sqrt{|b_1|}} SNR_t, \quad (31)$$

which was achieved when $\alpha' = |b_1|$.

Using (29) and (31), we have

$$\frac{\max_{\alpha}(SNR_{LCT}^{3dB})}{\max_{\alpha'}(SNR_{STFT}^{3dB})} = \sqrt{\frac{b_1}{\frac{a}{b} + b_1}}, \quad (32)$$

which means when $\alpha = |a/b + b_1|$, the 3dB SNR in the LCT domain can achieve a higher maximum than in the STFT domain, as long as $\min(0, 2b_1) < -a/b < \max(0, 2b_1)$. When $-a/b = b_1$, the ratio in (32) becomes infinity.

Using (30) and (31), we have

$$\frac{\max_A(SNR_{LCT}^{3dB})}{\max_{\alpha'}(SNR_{STFT}^{3dB})} = \sqrt{\frac{2|b_1|}{\alpha}}, \quad (33)$$

which means when $-a/b = b_1$, the 3dB SNR in the LCT domain can achieve a higher maximum than in the STFT domain, as long as $\alpha < 2|b_1|$. When $\alpha = 0$, the ratio in (33) also becomes infinity.

The windowed version of the WVD, which called pseudo-WVD (PWVD), is defined as [1]

$$PWVD_f(t, \omega) = \int f\left(t + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) h(\tau) e^{-j\omega\tau} d\tau. \quad (34)$$

According to [14], the 3dB SNR in the pseudo-WVD domain is

$$SNR_{PWVD}^{3dB} = 0.8\sqrt{2}(\pi/\alpha)^{1/4} SNR_t. \quad (35)$$

Using (30) and (35), we have

$$\frac{\max_A(SNR_{LCT}^{3dB})}{SNR_{PWVD}^{3dB}} = \sqrt{2}(\pi/\alpha)^{1/4}, \quad (36)$$

which means when $-a/b = b_1$, the 3dB SNR in the LCT domain can achieve a higher maximum than in the PWVD domain, as long as $\alpha < 4\pi$. When $\alpha = 0$, the ratio in (36) also becomes infinity.

III. SIMULATIONS

After theoretical analysis, some simulations are presented to confirm the theoretical results. In these simulations, we used an LFM Signal $s(t) = \exp[j(2t + 8t^2)]$ with noise $\eta(t)$, and $SNR_t = -3$ dB. Time t varies from -4 s to 4 s and the sampling rate is 50 HZ. Fig.1 shows the relationship between SNR_{LCT}^{3dB} and LCT parameter $A = (a, b, c, d)$. For a given $b = 1$ and a given $d = 1$, the result is obtained by changing a from -36 to -4. We can see the simulation result matches the theoretical result in (29), and the maximum 3dB SNR is obtained when $-a/b = 16$, which indicates when the opposite ratio between the LCT parameters a and b matches the signal chirp rate b_1 , the 3dB SNR in the LCT domain can reaches its maximum value.

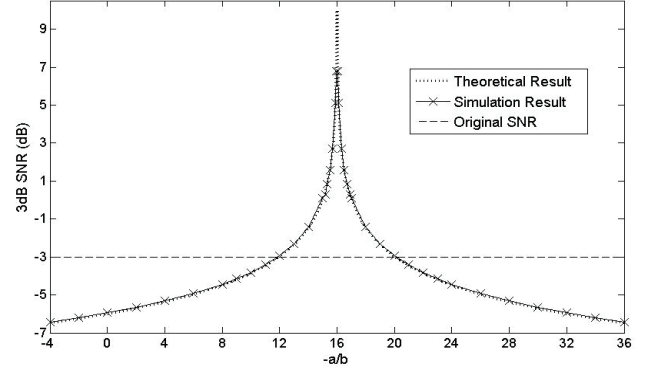


Fig. 1. The SNR_{LCT}^{3dB} versus $(-a/b)$

Fig.2 shows the relationship between SNR_{LCT}^{3dB} and window parameter α . For a given $A = (-16, 1, -17, 1)$, the result is obtained by changing α from 0.1 to 11. We can see the simulation result matches the theoretical result in (30).

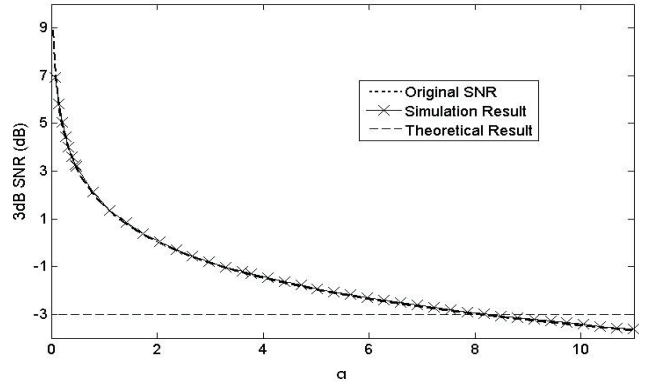


Fig. 2. The SNR_{LCT}^{3dB} versus α

Fig.3 presents the ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} with $\alpha = |a/b + b_1|$ and $\alpha' = |b_1|$. For a given $b = 1$ and a given $d = 1$, the result is also obtained by changing a from -36 to -4. This figure illustrates that when the opposite ratio between the LCT parameters a and b matches the signal chirp rate b_1 , the maximum ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} can be achieved. In this figure, the simulation result matches the theoretical result in (32).

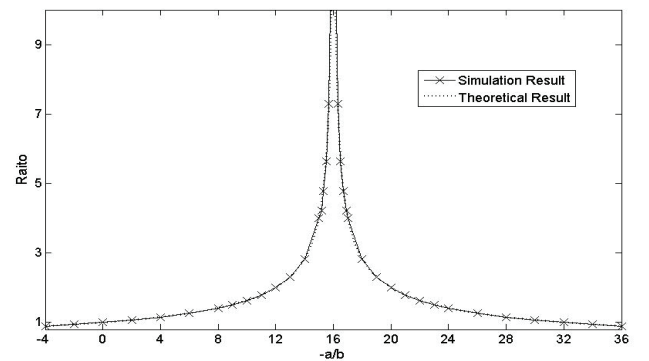


Fig. 3. The ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} versus $(-a/b)$

Fig.4 presents the ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} with a given $A = (-16, 1, -17, 1)$ and $\alpha' = |b_1|$, this

result is obtained by changing α from 0.1 to 11. This figure illustrates that when the LCT parameter A is fixed, the ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} becomes larger when the absolute value of the window parameter α becomes smaller. In this figure, the simulation result matches the theoretical result in (33).

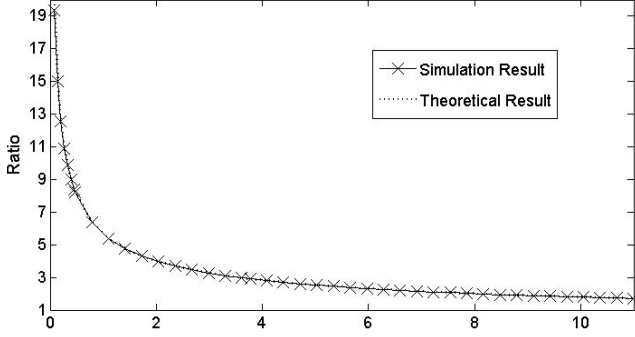


Fig. 4. The ratio between SNR_{LCT}^{3dB} and SNR_{STFT}^{3dB} versus α

Fig.5 presents the ratio between SNR_{LCT}^{3dB} and SNR_{PWVD}^{3dB} with a given $A = (-16, 1, -17, 1)$. This result is obtained by changing α from 0.1 to 14. This figure illustrates that when the LCT parameter A is fixed, the ratio between SNR_{LCT}^{3dB} and SNR_{PWVD}^{3dB} becomes larger when the absolute value of the window parameter α becomes smaller. In this figure, the simulation result matches the theoretical result in (36).

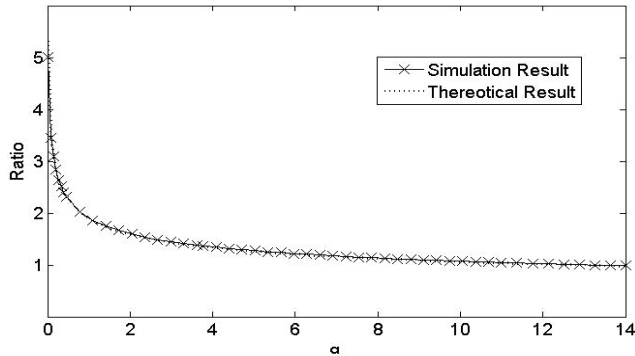


Fig. 5. The ratio between SNR_{LCT}^{3dB} and SNR_{PWVD}^{3dB} versus α

IV. CONCLUSION

In this paper, based on the relationship between the LCT and the WVD we proved, we theoretically got the relationship between the 3dB SNR in the LCT domain with Gaussian windows for LFM signals and the original SNR in the time domain for these signals. Then we found the condition which made the 3dB SNR obtain its

maximum value. We also compared the maximum 3dB SNR in the LCT domain with the SNRs in time domain, the STFT domain and the PWVD domain, respectively, to prove the SNR of LFM signals in the LCT domain can achieve a significantly higher level than the SNRs in the other three domains. At the end of this paper, some simulation results are presented to confirm our theoretical result.

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